Departamento de Ciências e Tecnologias da Informação

MESTRADO EM ENGENHARIA DE TELECOMUNICAÇÕES E INFORMÁTICA /
MESTRADO EM ENGENHARIA INFORMÁTICA

PROPOSTA DE DISSERTAÇÃO

Número da Proposta *(a preencher pelos serviços)*:

Título: Efficient Data Storage Using Programmable Containers in Edge Computing

Nome do orientador: José Moura

Nome do co-orientador: Pedro Ramos

Acompanhante de empresa

Nome:

Email:	**Telefone:**

Enquadramento *(tema, área científica, projecto, etc.; limitar a descrição a 1000 caracteres)*

The rise of Edge Computing and Inernet of Things (IoT) has introduced increasing demands for scalable and efficient data management. Ubiquitous access to modern services requires lightweight, portable execution environments that allow service deployment closer to data sources, enabling dynamic resource allocation and rapid scaling. In parallel, flexible data storage systems capable of handling both structured and unstructured metadata, such as, access frequency or spatio-temporal usage patterns, have become critical. However, coordinating the auto-scaling of services based on such meta-information remains a complex and unresolved challenge. This research aims to investigate programmable resource management strategies that adapt to data popularity trends, with the goal of optimizing service placement and data access in edge environments. The expected outcomes include contributions to real-time analytics, improved efficiency on resource usage, and enhanced decision-making across distributed computing environments.

Objectivos (limitar a descrição a 800 caracteres)

The generic objective of this MSc thesis is to investigate and develop an efficient approach for managing containerized resources in edge computing environments by leveraging programmable control and metadata-driven decision-making. The specific goals are:

1. To analyze the state of the art in edge computing, resource orchestration, and metadata-aware data management, identifying key challenges and research opportunities.

2. To design a programmable system architecture that supports dynamic scaling and efficient resource allocation based on spatio-temporal data popularity and other relevant metadata.

3. To implement a functional prototype that integrates containerized services with adaptive, metadata-informed resource management at the network edge.

4. To define and execute experimental scenarios that simulate realistic edge workloads and data usage patterns, supporting the complete evaluation of the proposed system.

5. To evaluate the system's performance using relevant metrics such as latency, scalability, and resource efficiency, and compare it against existing approaches or baselines.

6. To analyze and interpret the obtained results, discussing the strengths, limitations, and potential directions for future research or system optimization.

Descrição das actividades a desenvolver (limitar a descrição a 1000 caracteres):

The research will be conducted in the following phases:

1. Literature Review
1.1 Conduct a comprehensive review of the state-of-the-art in resource management for edge computing, with a focus on SDN, auto-scaling, containerization, and data popularity models.
1.2 Identify key concepts, frameworks, and research gaps relevant to programmable infrastructure and metadata-aware scaling.
1.3 Synthesize the findings to establish a strong theoretical foundation for the proposed approach.

2. Solution Design and Implementation
2.1 Design a system architecture that integrates programmable control, dynamic scaling, and container-based resource abstraction.
2.2 Specify system components, interaction flows, and decision-making algorithms, especially for handling spatio-

temporal metadata.
2.3 Implement a prototype system using appropriate development tools, emphasizing modularity and alignment with edge computing constraints.

3. Experimental Evaluation
3.1 Define evaluation objectives and formulate research hypotheses related to scalability, responsiveness, and data accessibility.
3.2 Design and configure experimental scenarios, including representative workloads, edge network topologies, and varying data popularity trends.
3.3 Use a document-oriented database to manage and query data with associated meta-information, supporting realistic edge data use cases.
3.4 Execute experiments and collect metrics such as response time, resource utilization, and data replication effectiveness.

4. Results Analysis and Validation
4.1 Analyze experimental data to assess the system's performance against defined hypotheses.
4.2 Interpret results in the context of the research objectives, highlighting the impact of metadata-aware scaling.
4.3 Compare outcomes with existing approaches or baselines to demonstrate improvements or trade-offs.
4.4 Discuss limitations and challenges encountered, proposing directions for refinement and future research.

Requisitos *(e.g. média, disciplinas concluídas)***:**

The student should have a solid background on programming, database management, and networked systems.

Resultado esperado *(protótipo, algoritmo, software, demonstração, …)***:**

Dissertation outcomes:

Full functional prototype

Master Dissertation

Scientific article, with enough quality to be published in a Conference or Journal.

Timeline:

Literature Review: 2 months

Algorithm Development: 3 months

Performance Evaluation: 2 months

Experimental Validation: 3 months

Dissertation Writing and Finalization: 2 months

URL da descrição detalhada da dissertação:

Localização da realização da dissertação:

IT-IUL, ISCTE-IUL research room; Student home.

Observações:

Some related references:

Chen, Peng, et al. "A Data Replication Placement Strategy for the Distributed Storage System in Cloud-Edge-Terminal Orchestrated Computing Environments." IEEE Internet of Things Journal (2025).

Fatemeh Karamimirazizi, Seyed Mahdi Jameii, and Amir Masoud Rahmani. 2024. Data Replication Methods in Cloud, Fog, and Edge Computing: A Systematic Literature Review. Wirel. Pers. Commun. 135, 1 (Mar 2024), 531–561. https://doi.org/10.1007/s11277-024-11082-7

E. Torabi, M. Ghobaei-Arani, and A. Shahidinejad, "Data replica placement approaches in fog computing: a review," Cluster Comput, 25, pp. 3561–3589, 2022, doi: 10.1007/s10586-022-03575-6

Distributed NoSQL Database Mongo, https://github.com/mongodb/mongo

SDN controller Ryu (Python), https://ryu-sdn.org/ Network emulator Mininet, https://mininet.org/ Software-based switch OpenvSwitch, https://www.openvswitch.org/ Containers management framework, https://www.docker.com/
Anexos